INCH-POUND

MIL-P-80276
29 June 1988

MILITARY SPECIFICATION

PRESS BRAKES, HYDRAULICALLY OPERATED, SHEET AND PLATE, COMPUTER NUMERICALLY CONTROLLED (CNC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers computer numerically controlled (CNC), hydraulically operated type press brakes with equipment and accessories specified herein.

1.2 Classification. The press brake shall be of the sizes listed in table I. The size to be furnished shall be as specified (see 6.2.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN. 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3441

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STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics Measurements of.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

29 CFR, 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of documents, Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods For Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1695).

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts, and Tongues.

ANSI B11.3 - Safety Requirements for the Construction, Care, and Use of Power Press Brakes.

ANSI NEMA MG 1 - Motors and Generators

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the International Organization for Standardization, C/O American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3363).

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-281 - Electrical and Construction Standards for Numerical Machine Control.

RS-232 - Interface between Data Terminal Equipment and Data Circuit-Terminating Equipment Employing Serial Binary Data Interchange.

(Application for copies should be addressed to Electronic Industries Association, Engineering Department, 2001 Eye Street, NW, Washington, DC 20006-1899.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, McLean, VA 22101-4269.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

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(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other information services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The press brake shall be new (not a protype), and one of the manufacturer's current models with an open throat type frame designed to perform forming, bending, flanging, and punching operations in accordance with the requirements herein. The press brake shall be designed for use in conjunction with the CNC system specified and shall include all components, parts, and features necessary to meet the performance requirements specified herein. The machine backgauge shall be CNC controlled. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Design, construction, and system of measurement. Unless otherwise required, either the inch/pound units or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all dimensions, measurements, sizes, and capacities are given in inch/pound units. The measurements may be converted to SI units through the use of the conversion factors and methods specified in FED- STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise required, either the inch/pound or the SI system of measurements shall be used to graduate measuring and indicating devices such as scales, depth stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1.1 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the inch/pound and the SI System of Measurements.

3.2.2 Reclaimed materials. The press brake may contain reclaimed materials provided such materials will not jeopardize the intended use, performance

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and design life of the machine. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The press brake and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be clearly marked and located conveniently to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the press brake that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of CFR 29, Chapter XVII, Part 1910 and ANSI B11.3 that are applicable to the machine itself. If a conflict arises between OSHA 29 CFR 1910 and B11.3, 29 CFR 1910 shall apply. The machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements shall be as specified in detail (see 6.2.1).

3.2.6 Mercury restriction. The press brake shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The press brake shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible. When specified (see 6.2.1), a centralized automatic lubrication system shall be provided.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

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3.3 Construction. The press brake shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine at its maximum bending capacity without permanent distortion. All weldments shall be stress relieved.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Gears. All gears and drive trains shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the inch/pound system or ISO 54 for the metric (SI) system. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirement shall be as specified.

3.4 Components. The press brake shall consist of not less than the frame, ram, hydraulic system, operating controls, computer numerical control, and the electrical system.

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3.4.1 Frame. The frame shall be of the open throat design consisting of a bed plate and side housings constructed of welded steel plate. Contoured stress points shall be utilized to avoid concentration of stresses at sharp corners. The frame shall have adequate strength and rigidity to support all components and withstand all forces encountered when bending sheet and plate of full capacity length and thickness in accordance with table I, for the machine size specified. The frame shall have mounting bolt holes for fastening the machine to the floor.

3.4.1.1 Bed plate. The bed plate shall be a solid steel plate joined to the side frames by either the interlocked and bolted construction method or welded to the side frames. When welds are used, the welds shall be stress relieved. The top of the bed plate shall have a flat machined surface and shall be slotted the full length to accommodate the manufacturer's standard die holder, filler blocks, or die tongues. The bed top width shall be not less than the dimensions shown in table I. Means shall be provided for adjusting and securely locking die or die holders to the bed. The bed plate shall have not less than four T-slots conforming to ANSI B5.1, running perpendicular to the tongue slot, for attaching short dies. Means shall be provided for attaching gauges and fixtures to the bed plate. When specified (see 6.2.1), the front and rear face of the bed plate shall be machined and provided with a method of mounting wide angle support brackets or flanges. The machine shall be furnished with a die holder or filler block and a punch with a mating die. The die holder or filler block shall be equal to the die surface length specified in table I. The type and size punches and dies furnished shall be as specified and fully described (see 6.2.1).

3.4.2 Ram. The ram shall be of solid steel plate, fitted with adjustable slides to work against steel or laminated self-lubricating non-metallic ways or antifriction rollers running on hardened ways. The lower edge of the ram shall be slotted to receive the manufacturer's standard die tongues. A positive clamping device shall be incorporated to prevent die movement under load. When specified (see 6.2.1), the front and rear face of the ram shall be machined and provided with a method of mounting wide angle brackets or flanges.

3.4.2.1 Ram or bed tilt adjustment. The press brake shall have means for tilting the ram or bed in either direction to facilitate taper or fade out work. The tilting device shall permit accurate adjustment of the desired angle between the ram and bed. An indicator graduated in increments of 0.001 inch shall be provided to accurately display the direction and amount of tilt. The tilt indicator shall be installed in a location readable from the operating position.

3.4.2.2 Ram leveling control. A ram leveling control shall be capable of maintaining the ram level to within ± 0.002 inch while bending sheet or plate of the full-length and thickness capacity as specified in table I.

3.4.3 Hydraulic system. The hydraulic system shall be provided with pumps, valves, piping, cylinders, and pressure controls. Over pressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean

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fluid. The hydraulic reservoir shall have means for determining fluid level. When specified (see 6.2.1), the hydraulic system shall meet or exceed the JIC H-1 standards. The hydraulic system shall be capable of producing full tonnage at any point in the ram stroke as required for the machine size specified in table I.

3.4.3.1 Ram hydraulic cylinder units. Unless otherwise required, the press brake shall have either one or more hydraulic units with mechanical linkage or two or more direct acting type hydraulic cylinder units for driving the ram. When only one type unit is acceptable, the particular hydraulic unit required shall be as specified (see 6.2.1). The cylinder unit shall be solid one-piece, tie rod, or bolted construction and shall be made of iron or steel. The cylinder unit shall be the double action piston type and the piston rod of each cylinder shall be sealed against leakage. The ram end of the piston rod on the direct acting cylinder units shall be provided with ball sockets or aligning joints with an adjustment for wear. The hydraulic units with the mechanical linkage shall also have provisions for adjustment and wear.

3.4.4 Operating controls. Unless a numerically controlled ram is supplied (see 3.4.5.2), all operating, stroking, speed, stroke length, and tonnage controls shall be provided. When used in conjunction with the hydraulic system, the controls shall provide not less than a high speed advance, slow working stroke, and a high speed return of the ram.

3.4.4.1 Stroking control. A stroking control shall be furnished to select the different modes of operation. The operating modes shall include not less than the following:

a. Off. The press brake shall be inoperative.
B. Single stroke. The ram shall complete only one stroke regardless of how long the actuating switches are energized.
C. Inch. Allow jogging of the ram at any point within the stroke.

3.4.4.2 Actuator controls. Unless otherwise specified (see 6.2.1), the actuator controls shall consist of both a remote foot switch and dual palm buttons. Machines having both controls shall have a means of selecting hand or foot operation. The palm buttons shall be connected so that concurrent pressure from both hands is required to actuate the press brake. The palm buttons may be located on the bed, ram, or on a free standing pedestal and shall be protected against unintentional actuation.

3.4.4.3 Ram speed controls. Unless otherwise specified, the ram speeds shall be the manufacturers standard speed rates. When other ram speed rates are required they shall be as specified (see 6.2.1). The ram speed control furnished shall enable the operator to select the advance, press, and return speed combinations best suited for the job. Necessary cams and switches shall be provided to set the desired speed change points throughout the stroke range.

3.4.4.4 Stroke length controls. Controls shall be provided to adjust the stroke length. The top stop and bottom depth stop settings shall be adjustable. The depth stop shall be adjustable within $\pm$ 0.001 inch.

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TABLE I. Sizes and capacities.

Size	Tonnage	Length of die surface (feet)	Bending capacity length x thickness	Distance between housings (inches)	Throat depth (inches)	Stroke length (inches)	Closed height (inches)	Open height (inches)	Bed top width (inches)	Motor horse-power
12-3	12	3	3' x 16 ga	25	6	1 1/2	7 1/2	9	2	2
12-4		4	4' x 18 ga	25	6	1 1/2	7 1/2	9	2	2
17-4	17	4	4' x 14 ga	36	6	2	6	8	2	2
35-4	35	4	4' x 12 ga	38						
35-6		6	6' x 14 ga	50	6	2	6	8	2	4
35-8		8	8' x 16 ga	62						
35-10		10	10' x 18 ga	98						
55-6	55	6	6' x 10 ga	50						
55-8		8	8' x 12 ga	74	6	2	6	8	2	6
55-10		10	10' x 14 ga	98						
75-6	75	6	6' x 3/16 in	50						
75-8		8	8' x 10 ga	74	6	3	6	9	2 1/2	6
75-10		10	10' x 12 ga	98						
100-6	100	6	6' x 3/16 in	50						
100-8		8	8' x 3/16 in	74	6	5	5	10	3	7
100-10		10	10' x 10 ga	98						
100-12		12	12' x 12 ga	122						
135-8	135	8	8' x 1/4 in	74						
135-10		10	10' x 3/16 in	98	7	5	5	10	3	10
135-12		12	12' x 10 ga	122						
135-14		14	14' x 10 ga	148						
150-8	150	8	8' x 1/4 in	72						
150-10		10	10' x 3/16 in	98						
150-12		12	12' x 3/16 in	122	7	7	5	12	4	10
150-14		14	14' x 10 ga	147						
150-16		16	16' x 10 ga	172						

NOTE: Requirements are not less than those shown.

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3.4.4.5 Tonnage control. The press brake shall be equipped with a conveniently located, adjustable tonnage control and a tonnage indicator that shows the load at all times.

3.4.5 Computer numerical control (CNC). The CNC shall be the solid state, soft wired, integrated circuit type numerical control constructed in accordance with EIA RS-281, and having an interface port complying with EIA RS-232. The control shall operate from a 120 volt, 60 Hz AC power supply. Line voltage variations of ± 10 percent of the specified voltage shall not adversely affect the operation of the control. Input snall be Manual Data Input (MDI) keyboards, standard 3 1/2 inch, floppy disc drive or magnetic cassette tape. Software resident in the control shall operate the machine from the part program, and shall perform the calculations for springback compensation, tonnage adjustment, and deflection compensation. The software shall provide diagnostic routines to assure that the machine, and control are operating properly. Malfunctions shall be displayed on the screen.

3.4.5.1 CNC backguage system. The backgauge shall have a single axis control to maintain horizontal backgauge positioning from the centerline of the die with an accuracy of ± 0.003 inch, repeatability of ± 0.002 inch, and resolution of not more than ± 0.001 inch. The backgauge horizontal travel shall be not less than 24 inches with control adjustment of not less than 6 inches. Manual vertical adjustment of the backgauge shall be not less than 6 inches. When specified (see 6.2.1), a two axes control shall be provided. The two axes control shall have the same features as the single axis control, in addition to a second axis to control vertical backgauge positioning.

3.4.5.2 CNC ram system. When specified (see 6.2.1), the CNC shall maintain ram positioning, velocity, and leveling. The control shall position the ram shut height within ± 0.002 inch, with repeatability of ± 0.001 inch. Resolution of the position feedback system used with the control shall be not more than 0.001 inch. The system shall be capable of controlling ram speeds, and speed change points.

3.4.5.3 Modes of operation. The CNC modes of operation shall include automatic, step, and manual. After initial set up, and the CNC in automatic mode, all machine operations shall be automatically controlled from the control's memory, cassette memory, or interface system. The step mode shall permit the operator to cycle each sequence of the part program individually. The manual mode shall permit the operator to control all backgauge movements from the MDI keyboard.

3.4.5.4 Control memory. The memory resident in the control shall have a capacity of not less than 20 programs with each program capable of not less than nine bend steps. Unless otherwise specified (see 6.2.1), controls having volatile part program memory shall have a battery backup capable of retaining part programs in memory for not less than 30 days.

3.4.5.5 Permanent memory. Unless otherwise specified (see 6.2.1), a permanent cassette memory, or floppy disc, shall be provided to permit infinite data storage and recall.

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3.4.5.6 Part program edit. A part program edit capability shall be provided to allow data insertion, deletion, or modification of any program in the part program memory. Data related to part program edit shall be input by MDI provided on the control panel. Edited program data entered into the program memory shall be automatically acted upon, in the sequence designated, whenever the part program is cycled.

3.4.6 Electrical system. The electrical system shall conform to NFPA 79. Each machine shall have a fused safety disconnect switch or circuit breaker. An identified terminal for grounding the machine when it is installed snall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating.

3.4.6.1 Primary input voltage. Unless otherwise specified (see 6.2.1), the machine shall be wired to draw all of its electrical power from a single 230/460 volt, 3-phase 60 Hz circuit. The machine shall be initially wired to operate from a 460 volt source.

3.4.6.2 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed, balanced, and permanently lubricated type. Alternating-current (AC) motors shall be designed to operate on 60-Hz. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall meet the requirements of ANSI/NEMA MG-1. Each motor shall have an identification plate to identify manufacturer, model, serial number, voltage, amperage, horsepower, phase, and frequency.

3.4.6.3 Solid state components. Solid state design shall be used throughout the press brake for electronic components.

3.4.6.4 Control circuit voltage. The control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 120 volts.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than those specified in table I for the machine size specified.

3.6 Performance. The CNC press brake shall be capable of bending steel sheet and plate of the capacities listed in table I for the machine size specified. The bending capacities are based upon steel sheet and plate with a tensile strength of not less than 60,000 pounds per square inch (psi). The machine shall be capable of performing full length bends and repetitions bends as follows:

3.6.1 Bending accuracy. The machine shall be capable of full length bends that are uniform throughout. The machine shall perform 90 degree bends and the angle shall be within $\pm$ 1 1/2 degree for the full length.

3.6.2 Repetitive bends. The machine shall be capable of making 18 repetitive air bends, which shall be compared, and each of the 18 pieces shall not vary more than $\pm$ 1 degree.

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3.6.3 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. Unless otherwise specified, all equipment normally furnished with the manufacturer's standard commercial machine shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.10 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.10.1 Lubrication chart or plate. A lubrication chart or plate shall be permanently and securely attached to each machine. The chart shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. A corrosion resistant nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Hour meter. Unless otherwise specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of

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not greater than one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.12 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

B. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

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4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The press brake shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than two hours. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Performance tests. The machine shall be tested for conformance with requirements of 3.6. The press brake shall then be fitted with a punch and the a V-die having an opening eight times the metal thickness. The inside radius shall be 5/32 of the V-die opening. The punch and V-die shall be properly designed to compensate for material spring back. The test sheet or plate shall be low carbon steel with a tensile strength of not less than 60,000 psi. The following tests shall be performed:

4.7.2.1 Full length bend test. The machine shall be set up for the MDI mode of operation. The test pieces shall be equal to the maximum capacity of the machine as specified in table I. Die and material specifications including springback allowance and angle data to form a 90 degree bend shall be programmed into the control. The test shall be accomplished by inserting a test piece between the dies and forming a 90 degree bend with one stroke of the ram. A total of 5 test bends shall be made. Ram leveling shall conform to 3.4.2.2.

4.7.2.2 CNC ram repetitive air bend test. When a CNC ram system is supplied the repetitive air bend test shall be performed by making air bends in sheet or plate of not less than 6 inches x 6 inches of the type and strength used in the full length bend test. The CNC shall be programmed to accomplish 30, 60, and 90 degree bends consecutively. The test pieces shall be positioned under each end of the ram in alignment with the end housings, and an air bend performed. The same air bend shall then be performed twice for each of the programmed degree settings. Two additional test pieces for each degree setting shall be bent individually at the left end of the ram with no material under the opposite end. The same air bend test shall be repeated on the right end of the ram.

4.7.2.3 Backguage accuracy test. A backguage accuracy test shall be performed. The test shall start with a zero reference to the centerline of the die. Six consecutive horizontal backgauge positions shall be programmed and executed by the MDI control. Each backgauge position shall be measured from the centerline of the die to assure positioning accuracy. The same horizontal backgauge positioning shall be repeated from the controller memory. The positioning and repeatability accuracies shall conform to 3.4.5.1.

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4.7.2.4 CNC ram accuracy test. When a CNC ram system is supplied, a test as determined by the manufacturer to demonstrate the accuracy of the ram shut height positioning, and repeatability under CNC shall be performed. The control shall be programmed for 6 different consecutive ram shut height positions. The ram shut height positioning accuracy shall be within ± 0.002 inch with a repeatability of ± 0.001 inch. The control shall be capable of maintaining ram level within ± 0.002 inch. The CNC shall be programmed to demonstrate not less than 3 ram speed change points enabling the operator to select the advance, press, and return speed combinations best suited for the bend being made.

4.7.3 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8, using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The numerically controlled press brake, covered by this specification is intended for use in forming, bending, flanging, multiple punching, and similar operations on sheet metal and plate.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2 and table 1).

c. First article, if required (see 3.1).

d. If machine design and construction is required in a specific dimensional system (inch/pound or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (inch/pound or SI), state required system (see 3.2.1.1).

f. Dual calibrations (inch/pound or SI), if required (see 3.2.1.1.1).

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g. Additional safety and health requirements, if required (see 3.2.5).

h. Centralized automatic lubrication system, if required (see 3.2.9).

i. Painting, if different (see 3.3.5).

j. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility requirement (see 3.3.8).

k. Machined bed, if required (see 3.4.1.1).

l. Type of die and punch to be furnished, fully describe (see 3.4.1.1).

m. Machined ram, if required (see 3.4.2).

n. Hydraulics conforming to JIC H-1, if required (see 3.4.3).

o. If the hydraulic unit(S) is to be a specific type, state required type (see 3.4.3.1).

p. Actuator controls, if different (see 3.4.4.2).

q. Ram speed combinations, if different (see 3.4.4.3).

r. Two axes backgauge, if required (see 3.4.5.1).

s. Specify and fully describe, CNC ram, if required (see 3.4.5.2).

t. Volatile memory battery backup, if different (see 3.4.5.4).

u. Permanent memory, if different (see 3.4.5.5).

v. Primary input voltage, if different (see 3.4.6.1).

w. Motor enclosure, if different (see 3.4.6.2).

x. Size and capacity, if different (see 3.5).

y. Additional performance features, if required (see 3.6.3).

z. Standard equipment, if different (see 3.7).

aa. Specify and fully describe optional equipment, if required (see 3.8).

bb. Specify applicable repair parts, if required (see 3.9).

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cc. Hour meter, if different (see 3.11).

dd. First article inspection, if different (see 4.4).

ee. Quality conformance inspection, if different (see 4.5).

ff. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level of preservation and packing required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Subject term (key word) listing.

Bending
Brake, press, forming, bending
Forming
Hydraulic press
Machine tools
Metal working

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activity:
Air Force - 84

User activities:
Army - AR
Navy - YD

Preparing activity:
DLA - IP

(Project 3441-0124)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-80276	Press Brakes, Hydraulically Operated, Sheet and Plate, Computer Numerically Controlled (CNC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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